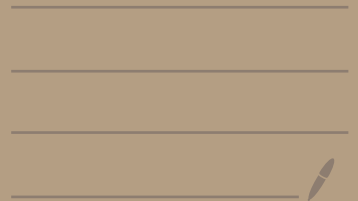


Lecture 32

Stability

- background
- Kelvin - Helmholtz



No class Wednesday 4/23.

Stability / Instability

- Most of the flows we've studied change when subjected to perturbation — often change a lot
 - infinitesimal : amplitude $\rightarrow 0$ but grow exponentially by linear mechanism $t \rightarrow \infty$
 - small but finite : amplify algebraically for finite t
 $\rightarrow$ can become large enough by linear mechanisms to trigger nonlinearity (and turbulence)

NOTE = any real flow has perturbations
... vibrations, thermal "noise"

Instability Analysis

- theory of perturbations (usually linear)
- questions
 - what grows? - "slope"/character of disturbances that grow
 - how fast? - important or not?
overwhelmed by turbulence?
- can be important
 - design of some air foils
 - heat transfer in hypersonic
- Turbulence is often the outcome
deterministic, linear amplification $\rightarrow$ chaotic, nonlinear
turbulence

- Boundary Layer Stability

- important, analytically intricate
- often relies on semi-analytic solutions

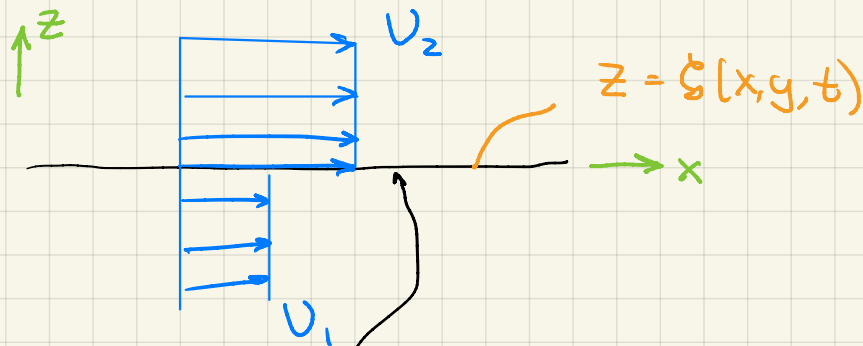
- Plan

- do a stability problem in full detail
 - to show how it "works"
 - free shear flow — Kelvin-Helmholtz (thin layer)
- derive main linear equations for BL stability
 - Orr-Sommerfeld equation
- discuss
 - equation + properties
 - solutions (quoted)
 - $\mu = 0$
 - how $\mu > 0$ can be destabilizing

- indifference Reynolds # and experiments
 ↳ below which perturbations decay

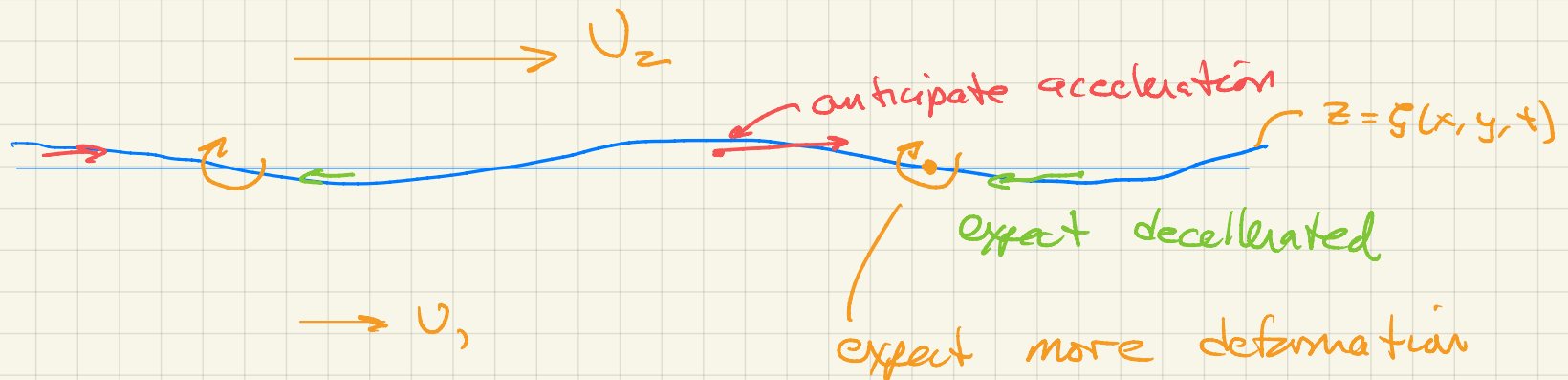
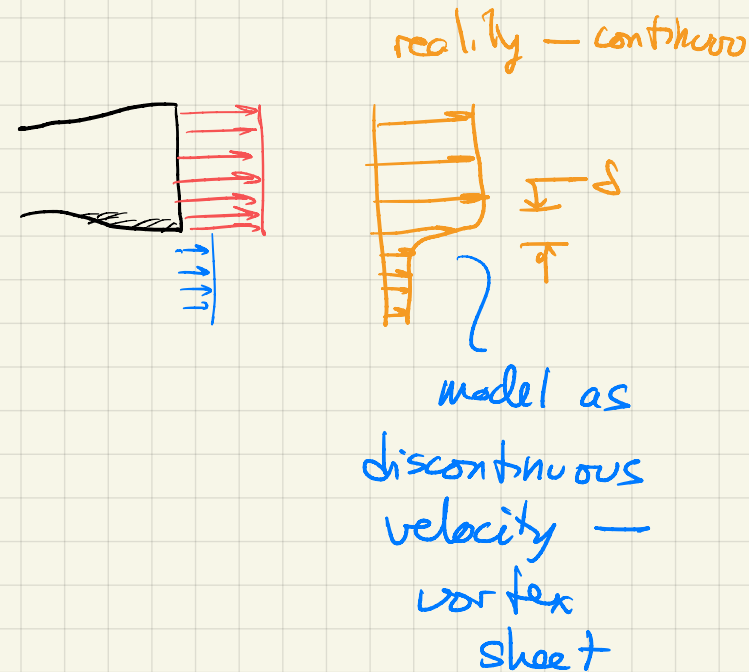
- transient growth finite time instability of
 nominally $t \rightarrow \infty$ stable flows

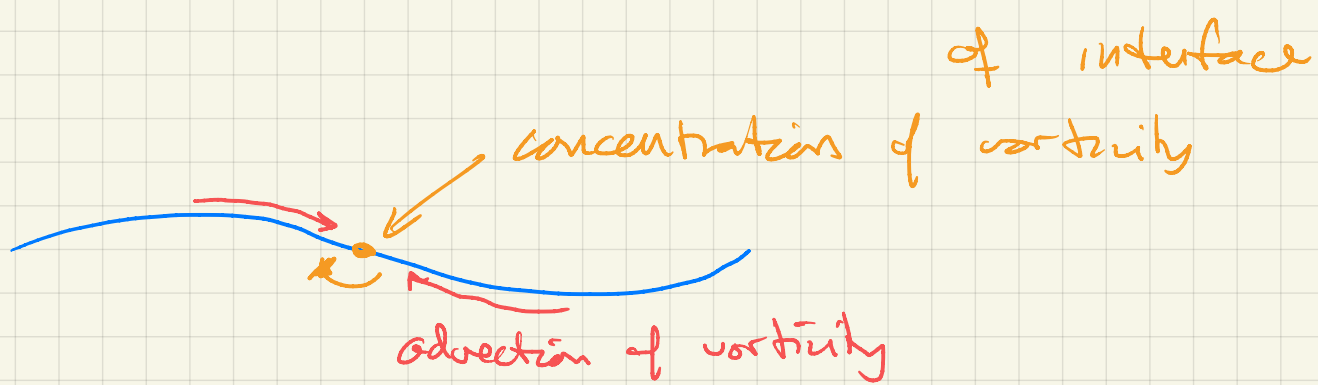
Kelvin-Helmholtz Instability



inviscid

what happens if interface is
perturbed?





incompressible

take $\underline{u} = \nabla \phi$
velocity potential

$$\phi = \begin{cases} \phi_1 & z < \zeta \\ \phi_2 & z > \zeta \end{cases}$$

GOV Eq

$$\nabla \cdot \underline{u} = 0 \Rightarrow$$

$$\begin{cases} \nabla^2 \phi_1 = 0 & z > \zeta \\ \nabla^2 \phi_2 = 0 & z < \zeta \end{cases}$$

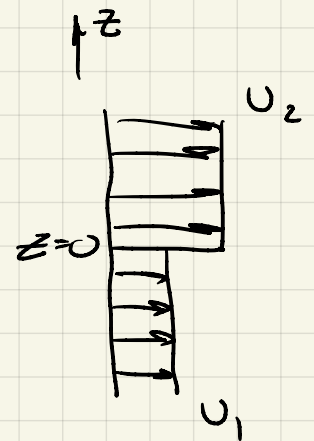
BC

• $u \rightarrow U_1$ for $z \rightarrow -\infty$

so $\nabla \phi_1 \rightarrow \underline{U}_1 = (U_1, 0, 0)$

• $u \rightarrow U_2$ for $z \rightarrow \infty$

so $\nabla \phi_2 \rightarrow \underline{U}_2 = (U_2, 0, 0)$



- interface in fluid "material"

define $f(x, y, z, t) = S(x, y, t) - z = 0$
 on interface

$$\frac{Df}{Dt} = 0$$

advection

$$\frac{\partial S}{\partial t} + u \frac{\partial S}{\partial x} + v \frac{\partial S}{\partial y} + w \frac{\partial (-z)}{\partial z} = 0$$

$\frac{\partial S}{\partial x}$ $\frac{\partial S}{\partial y}$ $\frac{\partial (-z)}{\partial z} = -1$

KINEMATIC BC

$$\begin{aligned} \frac{\partial \phi_1}{\partial z} &= \frac{\partial S}{\partial t} + \frac{\partial \phi_1}{\partial x} \frac{\partial S}{\partial x} + \frac{\partial \phi_1}{\partial y} \frac{\partial S}{\partial y} \\ \frac{\partial \phi_2}{\partial z} &= \frac{\partial S}{\partial t} + \frac{\partial \phi_2}{\partial x} \frac{\partial S}{\partial x} + \frac{\partial \phi_2}{\partial y} \frac{\partial S}{\partial y} \end{aligned}$$

- interface has no mass — stresses on it must balance

inviscid — $\underline{\underline{\sigma}} = -p \underline{\underline{S}}$ (only pressure)

unsteady Bernoulli: $\frac{1}{2} |u|^2 + \frac{p}{\rho} + \frac{\partial \phi}{\partial t} = C$

$$P_1 = \tilde{C}_1 - \frac{1}{2} S |\nabla \phi_1|^2 - S \frac{\partial \phi_1}{\partial t}$$

$$P_2 = \tilde{C}_2 - \frac{1}{2} S |\nabla \phi_2|^2 - S \frac{\partial \phi_2}{\partial t}$$

$$z \rightarrow +\infty \quad P_2 = \tilde{C}_2 - \frac{1}{2} S U_2^2$$

$$z \rightarrow -\infty \quad P_1 = \tilde{C}_1 - \frac{1}{2} S U_1^2$$

assume same P at $\pm \infty$

consistent w
 $W_{1,2} = 0$

$$C_2 - \frac{1}{2} U_2^2 = C_1 - \frac{1}{2} U_1^2$$

DYNAMIC
BC

$$z = \xi$$

$$C_2 - \frac{1}{2} |\nabla \phi_2|^2 - \frac{\partial \phi_2}{\partial t} = C_1 - \frac{1}{2} |\nabla \phi_1|^2 - \frac{\partial \phi_1}{\partial t}$$

Status: linear gov eq ($\nabla^2 \phi = 0$) with 2 nonlinear BC...

Linearize for small perturbations

$$\phi_1 = \underbrace{U_1 x}_{\text{base flow}} + \phi_1' \quad \frac{\partial U_1 x}{\partial x} = U_1$$

$$\phi_2 = U_2 x + \phi_2'$$

ϕ_2' perturbation, small, $O(\epsilon)$

$\rightarrow \epsilon$ small, $O(\epsilon)$

$$\underbrace{f(z=\epsilon)}_{\text{applying BC @ } z=\epsilon} = \underbrace{f(z=0)}_{\text{applying at } z=0} + \underbrace{\epsilon \frac{\partial f}{\partial z} \bigg|_{z=0}}_{\text{applying at } z=0} + O(\epsilon)^2$$

applying
BC @
 $z=\epsilon$

$\Leftrightarrow$

applying at
 $z=0$

To leading order
can apply BC
at $z=0$

• Linearize gov eq:

$$\nabla^2 \phi_1 = 0$$

$$\nabla^2 (U_1 x + \phi_1') = 0$$

$$\boxed{\begin{aligned} \nabla^2 \phi_1' &= 0 \\ \nabla^2 \phi_2' &= 0 \end{aligned}}$$

- Kinematic BC ($z=0$)

$$\frac{\partial \phi}{\partial z} = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial x} \frac{dx}{dz} + \frac{\partial \phi}{\partial y} \frac{dy}{dz} \quad (\text{from above})$$

$$\frac{f_e}{2} (\phi' + \phi'') = \frac{f_e}{2} (\phi' + \phi'') + \frac{x_e}{2} \frac{f_e}{x_e} (\phi' + \phi'') + \frac{f_e}{2} (\phi' + \phi'')$$

$$\frac{\partial \phi'}{\partial z} = \frac{\partial \phi'}{\partial t} + v \frac{\partial \phi'}{\partial x} + O(\varepsilon^2)$$

$$\frac{\partial \phi'}{\partial z} = \frac{\partial \phi_1'}{\partial z} + v_1 \frac{\partial \phi}{\partial x}$$

$$\frac{\partial \phi_2'}{\partial z} = \frac{\partial \phi_2'}{\partial z} + v_2 \frac{\partial \phi}{\partial x}$$

• dynamic BC

$$\frac{\partial \phi_1'}{\partial t} = O(\epsilon)$$

$$C_1 - \frac{1}{2} |\nabla \phi_1|^2 - \frac{\partial \phi_1}{\partial t} = C_2 - \frac{1}{2} |\nabla \phi_2|^2 - \frac{\partial \phi_2}{\partial t}$$

$$\uparrow$$

$$|\nabla (U_1 + \phi_1')|^2$$

$$(U_1 + \frac{\partial \phi_1'}{\partial x})^2 + \left(\frac{\partial \phi_1'}{\partial y}\right)^2 + \left(\frac{\partial \phi_1'}{\partial z}\right)^2$$

$$U_1^2 + 2U_1 \frac{\partial \phi_1'}{\partial x} + \left(\frac{\partial \phi_1'}{\partial x}\right)^2$$

$\sim 1 \quad \sim \epsilon \quad \sim \epsilon^2$

$$\nabla \phi_1' \rightarrow 0 \text{ at } z \rightarrow \infty$$

$$\nabla \phi_2' \rightarrow 0 \text{ at } z \rightarrow \infty$$

$\rightarrow O(1)$:

$$C_1 - \frac{1}{2} U_1^2 = C_2 - \frac{1}{2} U_2^2$$

✓ P balance
 $\pm \infty$

$\rightarrow O(\epsilon)$

$$U_1 \frac{\partial \phi_1'}{\partial x} + \frac{\partial \phi_1'}{\partial t} = U_2 \frac{\partial \phi_2'}{\partial x} + \frac{\partial \phi_2'}{\partial t}$$

take solutions of form